

1	Objective
2	Lab Environment
3	Network Configuration
4	Connectivity Verification
5	SSH Service Verification
6	Port Enumeration
7	SSH Login Attempts
8	Administrative Access Verification
9	User Enumeration and Account Review
10	User Creation Activity
11	Failed Authentication Analysis
12	Successful Authentication Analysis
13	Privileged Activity Analysis
14	Authentication Log Correlation
15	Findings
16	Conclusion

Linux SSH Authentication & Log Analysis

1. Objective

The objective of this lab was to configure SSH communication between two Linux systems, validate network connectivity, analyze authentication events, and investigate SSH-related activities through system logs. The exercise focused on understanding how successful logins, failed login attempts, user management actions, and privileged operations are recorded within Linux authentication logs.

2. Lab Environment

Component	Details
Attacker System	Kali Linux
Target System	Ubuntu Server
Network Type	VirtualBox Internal Network
Service	OpenSSH Server
Log Source	/var/log/auth.log

3. Network Configuration

Both virtual machines were connected to the same **Internal Network** within VirtualBox.

IP Address Assignment

System	IP Address
Kali Linux	192.168.56.10
Ubuntu Server	192.168.56.11

```
(shivangi@kali)-[~]
$ sudo ip addr add 192.168.56.10/24 dev eth0

(shivangi@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.10/24 scope global eth0
       valid_lft forever preferred_lft forever
```

```
vboxuser@ubuntu:~$ sudo ip addr add 192.168.56.11/24 dev enp0s3
[sudo: authenticate] Password:
vboxuser@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:77:08:17 brd ff:ff:ff:ff:ff:ff
    altname enx080027770817
    inet 192.168.56.11/24 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe77:817/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Observation

The internal network configuration isolated communication between the virtual machines and allowed controlled testing of SSH connectivity without external network exposure.

4. Connectivity Verification

Connectivity was verified using ICMP echo requests.

Command Used

- ping 192.168.56.11

```
(shivangi@kali)~$ ping 192.168.56.11
PING 192.168.56.11 (192.168.56.11) 56(84) bytes of data.
64 bytes from 192.168.56.11: icmp_seq=1 ttl=64 time=2.42 ms
64 bytes from 192.168.56.11: icmp_seq=2 ttl=64 time=3.39 ms
64 bytes from 192.168.56.11: icmp_seq=3 ttl=64 time=2.70 ms
64 bytes from 192.168.56.11: icmp_seq=4 ttl=64 time=2.67 ms
64 bytes from 192.168.56.11: icmp_seq=5 ttl=64 time=3.34 ms
64 bytes from 192.168.56.11: icmp_seq=6 ttl=64 time=2.63 ms
^C
— 192.168.56.11 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5036ms
rtt min/avg/max/mdev = 2.424/2.858/3.389/0.368 ms
```

Observation

Successful replies were received from the Ubuntu system.

Analysis

The results confirmed Layer 3 connectivity between both hosts and verified that network communication was functioning correctly before initiating SSH-based activities.

5. SSH Service Verification

The OpenSSH service was validated on the Ubuntu system.

Commands Used

- `sudo systemctl status ssh`
- `sudo systemctl start ssh`

```
vboxuser@ubuntu:/$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
            Docs: man:sshd(8)
                 man:sshd_config(5)
vboxuser@ubuntu:/$ sudo systemctl start ssh
vboxuser@ubuntu:/$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-06-19 17:28:59 UTC; 3s ago
     Invocation: f644e50878aa4488aaecc0be413bf18c
TriggeredBy: ● ssh.socket
            Docs: man:sshd(8)
                 man:sshd_config(5)
   Process: 3140 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 3143 (sshd)
    Tasks: 1 (limit: 1888)
   Memory: 1.3M (peak: 1.9M)
      CPU: 45ms
   CGroup: /system.slice/ssh.service
           └─3143 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 19 17:28:59 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 19 17:28:59 ubuntu sshd[3143]: Server listening on 0.0.0.0 port 22.
Jun 19 17:28:59 ubuntu sshd[3143]: Server listening on :: port 22.
Jun 19 17:28:59 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Observation

The SSH service was found running successfully and listening on port 22.

Analysis

Verifying the service ensured that the target system was prepared to accept incoming SSH connections.

Attacking System (kali)

6. Port Enumeration

A port scan was conducted from Kali Linux.

Command Used

- nmap 192.168.56.11

```
(shivangi@kali)-[~]  
$ nmap 192.168.56.11  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-19 13:56 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.  
Nmap scan report for 192.168.56.11  
Host is up (0.0019s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: 08:00:27:77:08:17 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.31 seconds
```

Observation

Port 22/TCP was identified as open.

Analysis

The scan confirmed that the SSH service was reachable across the network and actively accepting connections.

7. SSH Login Attempts

```
(shivangi@kali)-[~]
$ ssh vboxuser@192.168.56.11
vboxuser@192.168.56.11's password:
Permission denied, please try again.
vboxuser@192.168.56.11's password:
Permission denied, please try again.
vboxuser@192.168.56.11's password:
vboxuser@192.168.56.11: Permission denied (publickey,password).

(shivangi@kali)-[~]
$ ssh vboxuser@192.168.56.11
vboxuser@192.168.56.11's password:
Permission denied, please try again.
vboxuser@192.168.56.11's password:
Permission denied, please try again.
vboxuser@192.168.56.11's password:
vboxuser@192.168.56.11: Permission denied (publickey,password).

(shivangi@kali)-[~]
$ ssh vboxuser@192.168.56.11
vboxuser@192.168.56.11's password:
Permission denied, please try again.
vboxuser@192.168.56.11's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Jun 19 06:32:24 PM UTC 2026

System load: 0.16           Memory usage: 16%      Processes:      111
Usage of /:  14.0% of 24.44GB Swap usage:  0%       Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Jun 19 17:59:13 2026 from 192.168.56.10
vboxuser@ubuntu:~$
```

Observation

The screenshot displays three distinct remote connection attempts via SSH from the attacker terminal (shivangi@kali) targeting vboxuser@192.168.56.11. The first two multi-attempt authentication sequences resulted in failure, throwing Permission denied (publickey, password) errors. The third attempt succeeded after one failure, successfully spawning an interactive remote bash shell (vboxuser@ubuntu:~\$). The system banner confirms the target is running Ubuntu 26.04 LTS and explicitly logs the incoming connection source IP as 192.168.56.10.

Analysis

This sequence demonstrates a successful credential-matching process following repeated authentication failures, which closely resembles either a password-guessing/brute-force attempt or a user correcting typos. Once authentication succeeds, the SSH server automatically generates system telemetry (load, memory usage, and disk space details) and populates the login session history. This transition from unauthorized to authorized state establishes immediate user visibility and remote operational control over the target operating system.

8. Administrative Access Verification

Administrative privileges were reviewed using Linux user and sudo commands.

Commands Used

- Whoami
- sudo -l
- hostname
- sudo apt update

```
vboxuser@ubuntu:~$ whoami
vboxuser
vboxuser@ubuntu:~$ sudo -l
[sudo: authenticate] Password:
User vboxuser may run the following commands on ubuntu:
(ALL : ALL) ALL
vboxuser@ubuntu:~$ hostname
Command 'hostname' not found, did you mean:
  command 'hostname' from deb hostname (3.25build1)
Try: sudo apt install <deb name>
vboxuser@ubuntu:~$ hostname
ubuntu
vboxuser@ubuntu:~$ sudo apt update
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Ign:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
```

Observation

The current logged-in user was identified as **vboxuser**. The `sudo -l` command confirmed that the account possessed administrative privileges and was permitted to execute commands as root. The system hostname was also verified. During the package update attempt, DNS resolution failures prevented the system from reaching Ubuntu repositories.

Analysis

Privilege verification is commonly performed during system administration and investigations to determine the level of access available to a user account. The failed package update highlighted a network or DNS configuration issue rather than an authentication problem.

Key Finding

The **vboxuser** account possessed sudo privileges, while package update operations failed due to hostname resolution issues affecting external repository access.

9. User Enumeration and Account Review

Local user accounts were reviewed through system files.

Command Used

- cat /etc/passwd

```
vboxuser@ubuntu:~$ sudo cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:997:997:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:996:996:System Message Bus:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:990:990:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:989:989:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
usbmux:x:100:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
pollinate:x:101:1::/var/cache/pollinate:/bin/false
polkitd:x:988:988:User for polkitd:/:/usr/sbin/nologin
syslog:x:102:103::/nonexistent:/usr/sbin/nologin
uuid:x:103:104::/run/uuid:/usr/sbin/nologin
tcpdump:x:986:986:tcpdump:/nonexistent:/usr/sbin/nologin
tss:x:985:985:tss user for tpm2:/:/usr/sbin/nologin
landscape:x:104:106::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:984:984:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
vboxuser:x:1000:1000:vboxuser:/home/vboxuser:/bin/bash
sshd:x:983:65534:sshd user:/run/sshd:/usr/sbin/nologin
```

Observation

The /etc/passwd file was reviewed to identify local user accounts present on the Ubuntu system. The output contained both system accounts and interactive user accounts.

Analysis

Reviewing user accounts provides visibility into:

- Interactive users
- Service accounts
- Disabled accounts
- System processes

This information is often reviewed when investigating account misuse or unauthorized account creation.

Key Findings

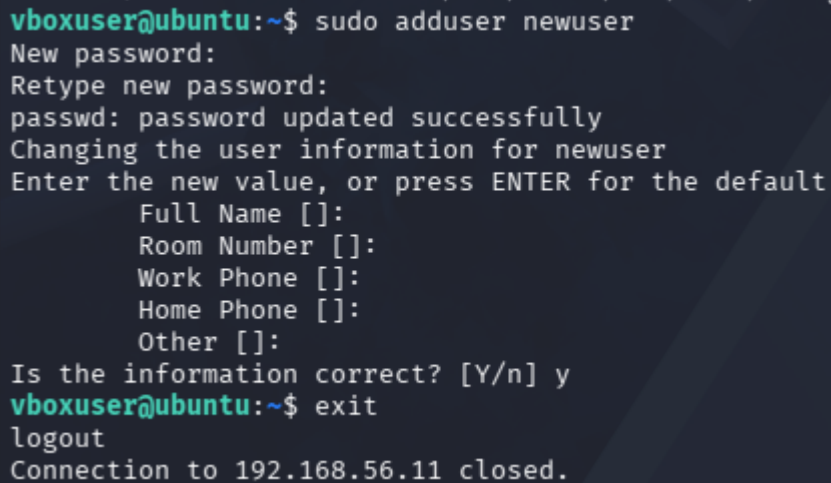
- vboxuser was identified as the primary local user account.
- sshd was present as a service account used by the OpenSSH daemon.

10. User Creation Activity

A new local user account was created into ubuntu system via ssh.

Command Used

- sudo adduser newuser



```
vboxuser@ubuntu:~$ sudo adduser newuser
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for newuser
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
vboxuser@ubuntu:~$ exit
logout
Connection to 192.168.56.11 closed.
```

Observation

The account creation process completed successfully.

Analysis

User creation events are important security activities because unauthorized account creation can indicate persistence attempts by an attacker.

Key Finding

The newly created account was successfully added and recorded within system logs.

Attacked System (Ubuntu) Logs

11. Failed Authentication Analysis

Multiple SSH login attempts were performed using incorrect credentials.

Evidence

Authentication failures were extracted directly from the system authorization log (/var/log/auth.log)

```
vboxuser@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log
[sudo: authenticate] Password:
2026-06-19T18:31:43.161195+00:00 ubuntu sshd-session[4035]: Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19T18:31:48.200108+00:00 ubuntu sshd-session[4035]: Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19T18:31:52.830122+00:00 ubuntu sshd-session[4035]: Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19T18:32:00.589768+00:00 ubuntu sshd-session[4040]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19T18:32:05.144256+00:00 ubuntu sshd-session[4040]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19T18:32:08.511806+00:00 ubuntu sshd-session[4040]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19T18:32:17.432875+00:00 ubuntu sshd-session[4045]: Failed password for vboxuser from 192.168.56.10 port 49780 ssh2
```

Analysis

The log entries captured during the investigation provide explicit metadata regarding the unauthorized connection traffic:

- **Target Account:** vboxuser
- **Source Attribution:** Traced to IP address 192.168.56.10 (Kali Linux attacker host).
- **Ports:** The connection rotated across multiple source ports (57198, 49760, 49780), indicating multiple distinct connection retries.
- **Session Processes:** Handled by sshd-session PIDs 4035, 4040, and 4045.

Security Context: Analyzing rapid, successive authentication failures within a concise timeframe is crucial during incident response. This behavior patterns authentic indicator trends for **brute-force activity**, **automated password guessing**, or an unauthorized user attempting credential stuffing.

Key Finding

The target system effectively enforced its authentication controls by rejecting invalid login attempts. The security subsystem properly audited the events, mapping the malicious footprint definitively back to the Kali Linux source node.

12. Successful Authentication Analysis

A successful remote SSH interactive login was established on the target system using valid account credentials.

Evidence

Authentication failures were extracted directly from the system authorization log (/var/log/auth.log)

```
vboxuser@ubuntu:~$ sudo grep "Accepted password" /var/log/auth.log
2026-06-19T18:32:24.321096+00:00 ubuntu sshd-session[4045]: Accepted password for vboxuser from 192.168.56.10 port 49780 ssh2
2026-06-20T17:29:39.121197+00:00 ubuntu sudo: vboxuser : TTY=/dev/tty1 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/grep Accepted password /var/log/auth.log
```

Analysis

The log entry validates the transition from an unauthenticated connection to an active session, providing explicit baseline metrics:

- **Validated Identity:** The authorization subsystem accepted credentials for user account vboxuser.
- **Attribution Mapping:** The connection successfully originated from IP address 192.168.56.10 (Kali Linux host) via source port 49780.
- **Session Tracking:** The connection was established through sshd-session process ID 4045 utilizing the ssh2 protocol.

Successful authentication records are frequently used during incident investigations to reconstruct user activity timelines.

Key Finding

The successful login occurred exactly 7 seconds after the final failed entry from port 49780. This provides an unbroken behavioral profile, confirming that the initial unauthorized noise was directly followed by authorized interactive console access.

13. Privileged Activity Analysis

An audit of administrative actions executed via sudo was conducted to map out behavioral patterns and system modifications performed during the active ssh session.

Evidence

Privileged lifecycle logs were extracted from /var/log/auth.log using the sudo keyword filter:

```
vboxuser@ubuntu:~$ sudo grep "sudo" /var/log/auth.log
2026-06-19T18:29:47.719413+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:30:34.266763+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:30:34.267579+00:00 ubuntu sudo: vboxuser : TTY=/dev/tty4 ; PWD=/ ; USER=root ; COMMAND=/usr/bin/systemctl start ssh
2026-06-19T18:30:34.306212+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:30:42.457849+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:30:42.458937+00:00 ubuntu sudo: vboxuser : TTY=/dev/tty4 ; PWD=/ ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
2026-06-19T18:30:42.488033+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:30:45.386480+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:30:45.404758+00:00 ubuntu sudo: vboxuser : TTY=/dev/pts/0 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/apt update
2026-06-19T18:30:59.050687+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:39:25.798138+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:39:25.798775+00:00 ubuntu sudo: vboxuser : TTY=/dev/pts/0 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/apt update
2026-06-19T18:39:36.614508+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:40:16.286153+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:40:16.293554+00:00 ubuntu sudo: vboxuser : TTY=/dev/pts/0 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
2026-06-19T18:40:26.569414+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:40:48.752339+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:40:48.754702+00:00 ubuntu sudo: vboxuser : TTY=/dev/pts/0 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/cat /etc/passwd
2026-06-19T18:40:48.783697+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-19T18:42:49.632729+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
2026-06-19T18:42:49.653977+00:00 ubuntu sudo: vboxuser : TTY=/dev/pts/0 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/sbin/adduser newuser
2026-06-19T18:43:06.567639+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
```

Analysis

The syslog payload provides comprehensive non-repudiation and structural oversight regarding administrative actions:

- **Accountability:** The operator vboxuser is explicitly named as the initiator of the privilege escalation requests.
- **Target Execution State:** All commands were processed under USER=root, granting absolute system access.
- **TTY/Context Transition:** Early commands were invoked via local terminal TTY=dev/tty1 (prior to SSH initiation), while later discovery and management commands were run via TTY=pts/0, confirming execution through the remote SSH tunnel.
- **Command Sequence Auditing:** The logs chronologically trace the operational footprint:
 1. Service activation and state monitoring (systemctl start/status ssh).
 2. Network package indexing requests (apt update).
 3. Sensitive configuration read operations (cat /etc/passwd).
 4. Persistent system modifications (adduser newuser).

Security Context: Continuous monitoring of sudo execution logs is essential for maintaining compliance baselines. It enables security teams to quickly distinguish routine administrative tasks from anomalous access patterns, lateral movement, or unauthorized account persistence.

Key Finding

Every elevated command executed during this session left a clear, tamper-evident audit trail within /var/log/auth.log. This log data maps the entire operational progression from the initial SSH daemon setup to target reconnaissance and user provisioning under a unified administrative profile.

14. Authentication Log Correlation

Events recorded throughout the investigation were correlated to reconstruct activity.

Activity Timeline

1. Internal network configured.
2. IP addresses assigned.
3. Connectivity verified through ICMP.
4. SSH service enabled and verified.
5. Port 22 discovered via Nmap.
6. Failed login attempts generated.
7. Successful authentication performed.
8. User accounts reviewed.
9. New user account created.
10. Administrative actions executed through sudo.
11. Authentication logs analyzed and correlated.

15. Findings

1. SSH service was successfully deployed and verified.
2. Network communication functioned correctly between both hosts.
3. Failed login attempts were recorded with source attribution.
4. Successful authentication events were logged accurately.
5. User enumeration activities were performed successfully.
6. User creation events were completed and documented.
7. Sudo operations generated detailed audit records.
8. Authentication logs provided clear visibility into user actions.

16. Conclusion

This lab successfully demonstrated the deployment, monitoring, and forensic auditing of an isolated network environment.

- **Network Isolation:** The VirtualBox internal network safely sandboxed all traffic, verified by successful internal pings and expected external update failures.
- **Authentication Auditing:** Analysis of `/var/log/auth.log` clearly distinguished between unauthorized entry patterns (failed password attempts from IP 192.168.56.10 across rotating ephemeral ports) and authorized access.
- **Administrative Accountability:** Every elevated action—including SSH activation, system reconnaissance (`cat /etc/passwd`), and account creation (`adduser`)—was captured with a non-repudiable timestamp under `sudo logs`.